filtering the screening dataset a pisum

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Setup

Packages required

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("ggrepel"))
```

“

Import helper functions

```
source("../broad_heritability_function.R")
source("../create_correlation_plot.R")
```

Broad sense heritability calculations

Raw dataset

```
data250pisum_00 <- read.csv("../../../../../01.data/2023_screen_specifics_251genotypes/01.acy
                            stringsAsFactors = FALSE,
                            sep = ",") %>%
  mutate(genotype = as.factor(genotype)) %>%
  mutate(session = as.factor(session))

res <- calculate_broad_heritability(data = data250pisum_00)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.171.
> - There are 251 genotypes in total.
> - There are 3027 measurements in total.

Missing data removed

```
data250pisum_01 <- data250pisum_00 %>% filter(!is.na(larvae_total_number))
data250pisum_01$genotype <- droplevels(data250pisum_01$genotype)

res2 <- calculate_broad_heritability(data = data250pisum_01)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.171.
> - There are 249 genotypes in total.
> - There are 2650 measurements in total.

```
create_correlation_plot(data = data250pisum_01)
```

Remove observations with no adult alive

We chose to remove all data from plants on which no adult aphid was found. This decision was made to get rid of larvae installation bias and only assess the reproduction ability of the aphid.

When the adult aphid dies, the number of larvae will drop but only because the adult stayed less longer on the plant and not because the plant was not suitable for the reproduction.

```
data250pisum_02<- data250pisum_01 %>% filter(adult_alive != "0")
data250pisum_02$genotype <- droplevels(data250pisum_02$genotype)

res3 <- calculate_broad_heritability(data = data250pisum_02)
```

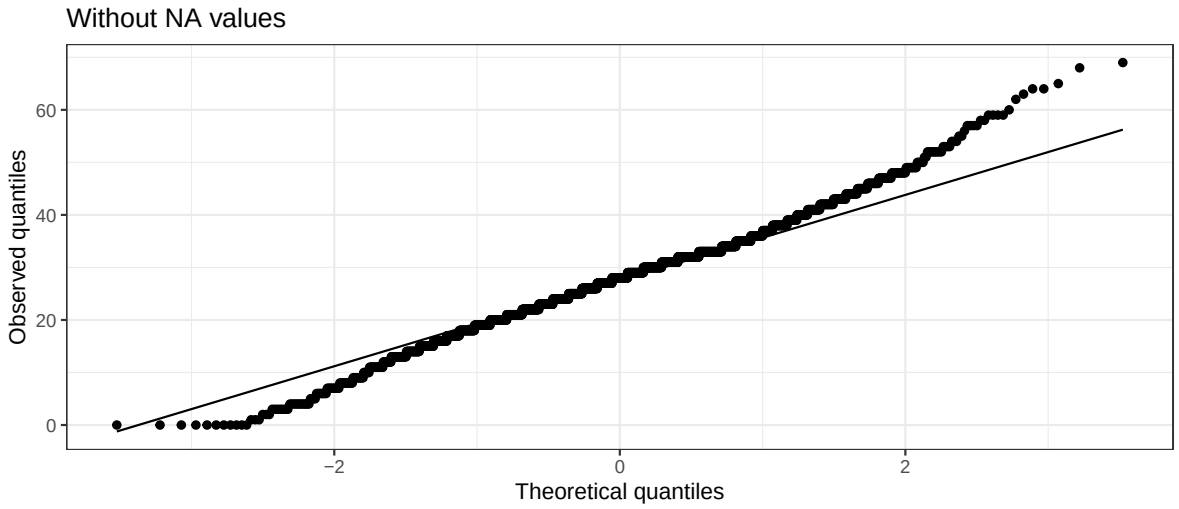
> Tip
>
> Conclusions:
> - Heritability is equal to 0.369.
> - There are 248 genotypes in total.
> - There are 2345 measurements in total.

This filtering led to the loss of 1 genotype but increased broad sense heritability up to 0.36

```
# Call the function to generate and display the correlation plot
create_correlation_plot(data250pisum_02,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

Distribution normality check.

```r
p1 <- data250pisum_02 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Without NA values") +
  theme_bw()
p1
```



Finding the optimal number of biological replicates

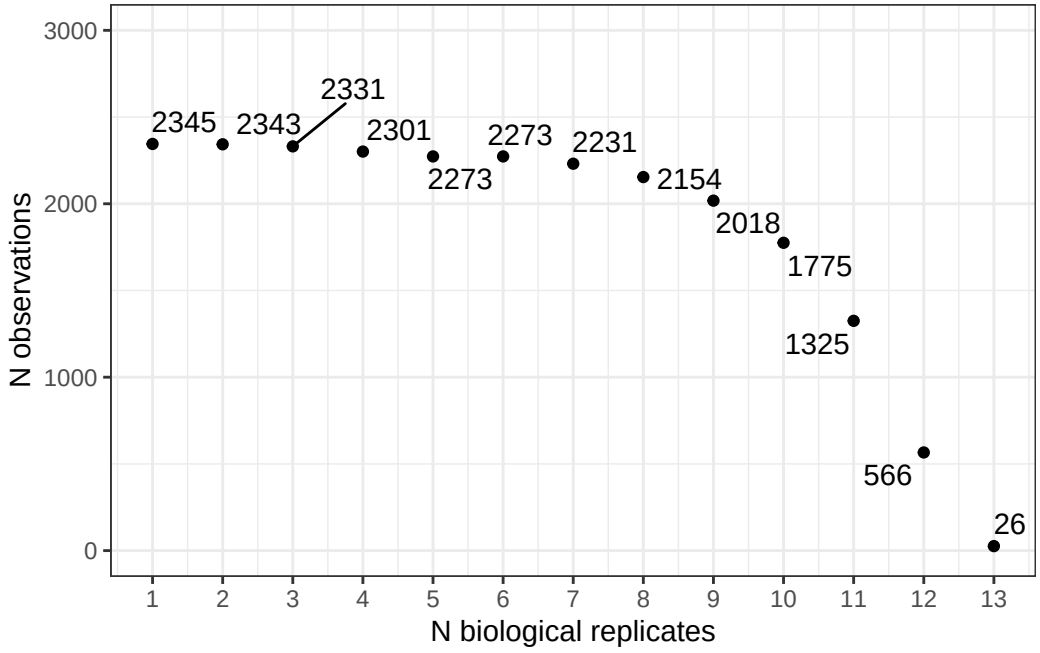
Now the question is: should we filter genotypes for which we have less than X biological repetitions?

Let's extract:
- The number of genotypes available for X number of biological replicates.
- The number of observations available for X number of biological replicates.
- The broad-sense heritability for X number of biological replicates.

```
boundary (singular) fit: see help('isSingular')
```

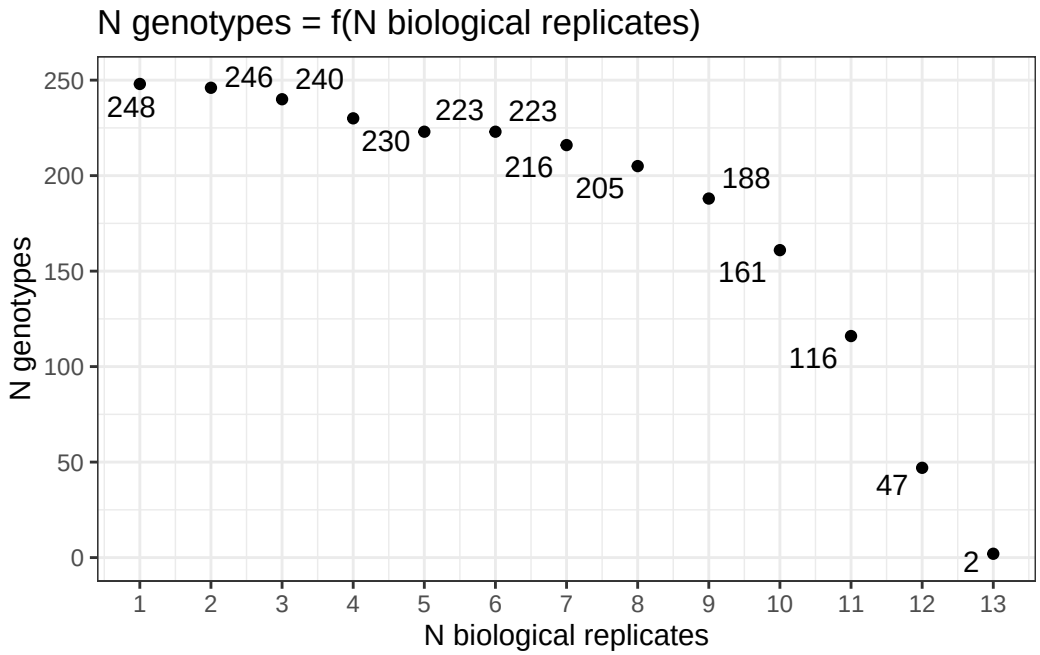
N observations = f(N biological replicates)

```
ggplot(dfb, aes(x = Nrep, y = Nob)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,3000)) +
  geom_text_repel(aes(label = Nob)) +
  theme_bw() +
  labs(x = "N biological replicates", y = "N observations")
```



N genotypes = f(N biological replicates)

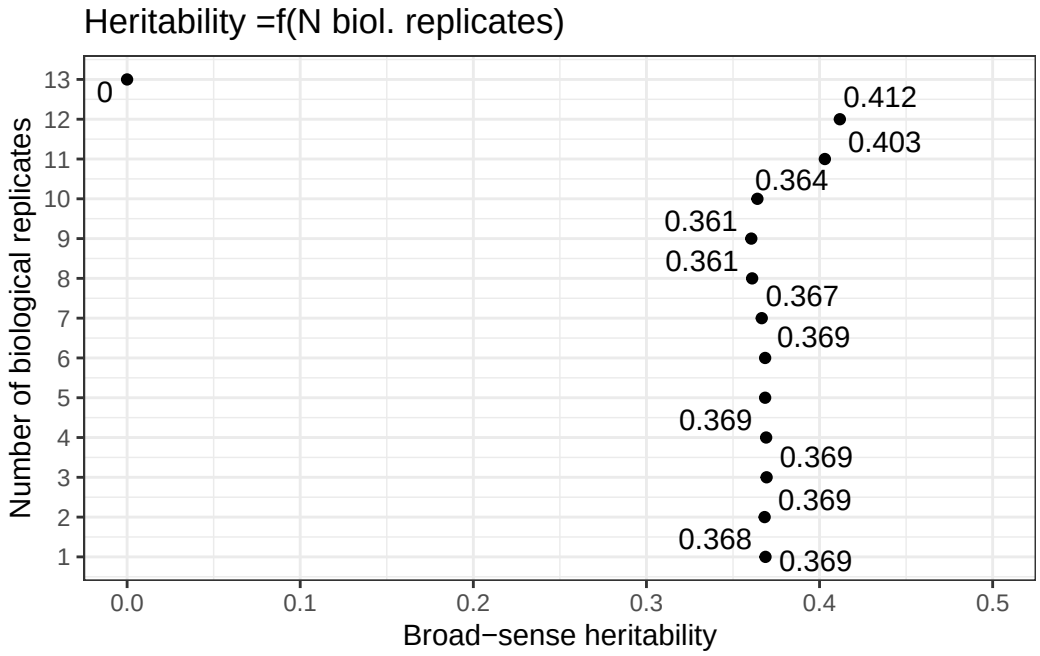
```
ggplot(dfb, aes(x = Nrep, y = Ngen)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,250)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = Ngen), arrow = T) +
  labs(x = "N biological replicates", y = "N genotypes") +
  ggtitle("N genotypes = f(N biological replicates)")
```



Heritability = f(N biological replicates)

```
ggplot(dfb, aes(x = H2, y = Nrep)) +
  geom_point() +
  scale_y_continuous(breaks=seq(1,13,1)) +
  scale_x_continuous(limits = c(0,0.5)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = round(H2,digits = 3))) +
```

```
  labs(y = "Number of biological replicates", x = "Broad-sense heritability") +
  ggtitle("Heritability =f(N biol. replicates)")
```



> **!** Important
>
> Based on these results, a threshold of >=6 biological replicates seems appropriate. That is keeping all genotypes for which at least 6 biological replicates are available.

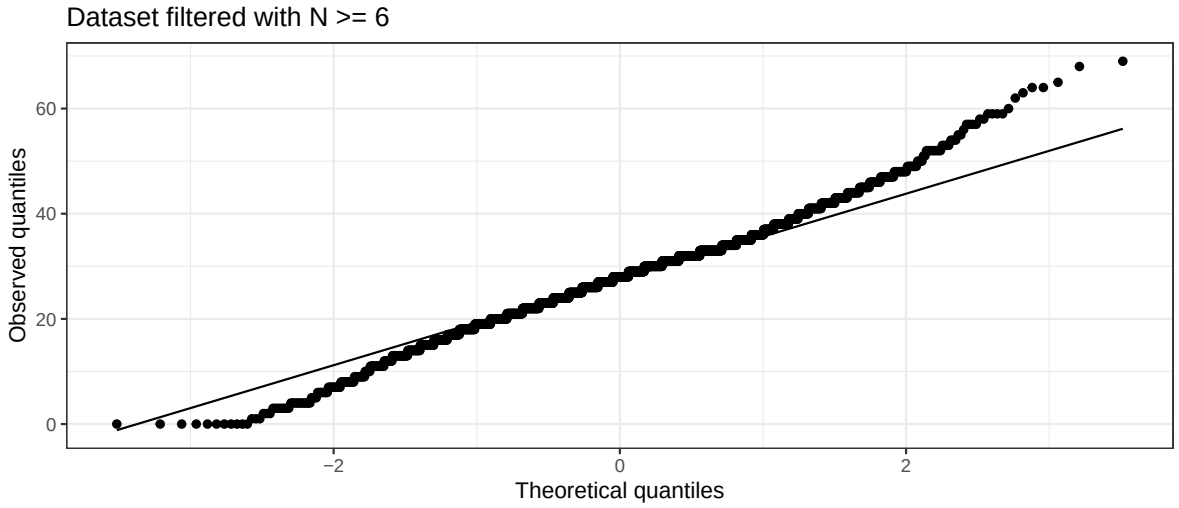
Final filtering: keep genotypes with at least 6 biological repetitions

```
data250pisum_04 <- data250pisum_02 %>%
  group_by(genotype) %>%           # Group by genotype
  filter(n() >= 6) %>%             # Keep only groups with 7 or more repetitions
  ungroup()
data250pisum_04$genotype <- droplevels(data250pisum_04$genotype)
res4 <- calculate_broad_heritability(data = data250pisum_04)
```

> **!** Important
>
> Conclusions:
> - Heritability is equal to 0.369.
> - There are 223 genotypes in total.
> - There are 2273 measurements in total.

Distribution normality check.

```
p <- data250pisum_04 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Dataset filtered with N >= 6") +
  theme_bw()
p
```



Write final dataset to .csv

```
write.csv(x = data250pisum_04,
          file = "../../../01_phenotipic_analysis/acyrthosiphon_pisum_filtered.csv",
          quote = F, row.names = F)
```